#5: 4abce, 20, 38a, 44

Homework 7, Math 3070 By: Tyler Trett

4: a) $\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 51.4 \pm 1.96\left(\frac{2.8}{\sqrt{25}}\right)$

$= \boxed{51.4 \pm 1.96(.56)}$

b) $\bar{x} = 51.4$, $\sigma = 2.8$, $n = 100$, $\alpha = .05$

$\bar{x} \pm Z_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right) = 51.4 \pm 1.96\left(\frac{2.8}{10}\right) = \boxed{51.4 \pm .5488}$

c) $\bar{x} \pm Z_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right) = 51.4 \pm 2.576\left(\frac{2.8}{\sqrt{100}}\right) = \boxed{51.4 \pm .7213}$

e) $n = \left(\frac{\sigma Z_{\alpha/2}}{E}\right)^2 = \left(\frac{2.8 \cdot 2.576}{.5}\right)^2 = \boxed{208.0656}$

20: a) $CI = \hat{p} \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = .53 \pm 2.58\sqrt{\frac{(.53)(1-.53)}{2343}}$

$\boxed{= .53 \pm .027}$

b) $n = \frac{4z^2 \hat{p}(1-\hat{p})}{w^2} = \frac{4 \cdot 2.58^2 \cdot .53(1-.53)}{(.05)^2}$

$= 2652.975 = \boxed{2653}$

38: a)

44: $n = 9$ $\sigma = 2.84$ $CI = 1 - \alpha = .95$ $\alpha = .05$

$DF = n - 1 = 9 - 1 = 8$

$X^2_{1-\alpha, n-1} = X^2_{0.025, 8} = 17.535$

$X^2_{1-\alpha, n-1} = X^2_{0.975, 8} = 2.180$

$$\frac{(9-1)(2.84)^2}{17.535} \le \sigma^2 \le \frac{(9-1)(2.84)^2}{2.180}$$

$3.6797772 \le \sigma^2 \le 29.59853$

$\sqrt{3.680} \le \sigma^2 \le \sqrt{29.599}$

$1.918 \le \sigma^2 \le 5.440$